

Curve Finance Review

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Abstract—This document provides a concise review of Curve Finance, focusing on its AMM models, governance structure, tokenomics, and risk profile. We analyze the StableSwap and CryptoSwap invariants, the veCRV voting system, and the crvUSD stablecoin with its LLAMMA liquidation mechanism. The study also examines Curve’s multichain deployment, security incidents, and economic dynamics, offering an integrated overview of the protocol’s design, strengths, and limitations within the broader DeFi ecosystem.

I. INTRODUCTION

Decentralized Finance (DeFi) has become one of the most active areas in blockchain, offering financial services without traditional intermediaries. Instead of relying on banks or centralized exchanges, DeFi protocols use smart contracts that allow users to trade, lend, borrow, and earn yield in a transparent and permissionless way [1]. Within this ecosystem, decentralized exchanges (DEXs) and automated market makers (AMMs) play a central role, enabling users to swap tokens through liquidity pools rather than order books. [2], [3]

Curve Finance is one of the most influential AMMs in DeFi. Launched by Michael Egorov in late 2019, Curve was designed to enable efficient trading between stablecoins and other pegged assets by minimizing slippage and reducing risks for liquidity providers. To achieve this, Curve introduced the StableSwap algorithm, which combines constant-product and constant-sum formulas to maintain stable prices and deep liquidity for assets that should remain close in value [4]. Over time, the protocol expanded its design to support more volatile assets through the CryptoSwap model [5], and later introduced crvUSD, a stablecoin based on a new liquidation mechanism called LLAMMA [6].

Curve also operates as a decentralized autonomous organization (DAO), where governance is managed through the CRV token and its vote-escrowed form, veCRV. This governance model aligns incentives by rewarding long-term commitment and giving users influence over token emissions, pool approvals, and protocol parameters [7], [8].

Because of its technical innovations, strong liquidity, and active governance system, Curve has become a key infrastructure component in DeFi. Its pools influence stablecoin pricing, liquidity flows, and the broader dynamics known as the “Curve Wars,” where protocols compete for voting power to direct CRV emissions [9]. However, the platform also faces important risks, including peg instability, oracle dependen-

cies, governance attacks, and smart-contract vulnerabilities, as demonstrated by past incidents.

II. BACKGROUND

A. Blockchain Fundamentals

1) *Distributed Ledgers*: Distributed Ledger Technology (DLT) is a generalized distributed protocol designed to securely record data across a decentralized network without relying on a central authority. It establishes trust among participants by distributing multiple copies of records, cryptographically linked to prevent tampering. Each record is stored in a linked manner, secured by cryptography, making it immutable and transparent across all nodes in the network. The main purpose of DLT is to address limitations of traditional distributed computing by providing transparency, security, and decentralization, enabling applications that extend beyond its origins in cryptocurrency. [10]

2) *Blocks, Transactions, and Addresses*: In Blockchain technology, data are recorded in a chronological chain of blocks, each containing a set of transactions, a unique hash of its contents, and the hash of the previous block, ensuring tamper-resistance through cryptographic linkage. Transactions represent the fundamental data units, typically including sender and recipient addresses and transferred values. Addresses act as unique identifiers in the network, allowing participants to securely sign and verify transactions using public-private key cryptography. Blocks are timestamped and linked, beginning with the genesis block, and any alteration in transaction data modifies the block’s hash, invalidating subsequent blocks. This layered structure makes altering historical records computationally infeasible, preserving the immutability essential to Blockchain’s trust model. [10]

3) *Consensus Mechanisms*: Consensus mechanisms are essential protocols that enable decentralized blockchain networks to agree on the validity and order of transactions without relying on a central authority. They ensure that all nodes share a consistent view of the ledger and prevent issues such as double spending or conflicting blocks. This need arises from the Byzantine Generals Problem, where nodes must reach agreement even when some may act maliciously or provide false information. Among the fundamental consensus mechanisms are Proof of Work (PoW) and Proof of Stake (PoS). PoW requires miners to solve complex cryptographic puzzles, and the first to find the solution earns the right to

add a new block. This method provides strong security and decentralization but suffers from high energy consumption and limited scalability. PoS, on the other hand, selects validators based on the amount of cryptocurrency they hold and commit as stake. This approach greatly reduces energy usage and can improve scalability, though it may introduce concerns about wealth concentration. Each consensus mechanism involves trade-offs across security, decentralization, throughput, and resource consumption. [11]

B. Smart Contracts Ecosystem

1) *Smart Contracts*: Curve Finance uses smart contracts, which are defined as self-verifying, self-executing, and tamper-resistant computer programs that run on a blockchain platform to automate transactions without relying on third parties. In these contracts, the terms of the agreement are directly written into code and executed when predefined conditions are met, allowing value, addresses, functions, and state to operate autonomously. Their automation eliminates the need for intermediaries, improves efficiency, and reduces the likelihood of disputes, while also enabling trusted interactions within decentralized and trustless environments. Smart contracts provide transparency through publicly verifiable operations recorded on the blockchain, ensure immutability by preventing modifications once deployed, maintain autonomy by functioning without human intervention, and enhance security through cryptographic guarantees and decentralized execution. These features enable real-time, secure, and transparent operations not only in DeFi, where they facilitate automated trading, lending, and borrowing, but also across broader sectors such as supply chain tracking, healthcare management, and voting systems. [12], [13]

2) *ERC-20 Tokens*: A token is a type of digital asset created through a smart contract. It allows developers to represent items that can be transferred and counted, such as digital or physical assets, shares, votes, memberships, or loyalty points. Because tokens depend on smart contracts instead of a blockchain's built-in currency system, anyone can create and distribute their own asset without building a new blockchain. This makes issuing tokens easier, quicker, and more affordable. The ERC-20 standard provides a common set of rules for these assets, ensuring consistent transfers, balance management, and compatibility across wallets, exchanges, and other applications. [14]

3) *Gas*: A gas represents the unit of computational cost required to execute operations within a blockchain network, such as Ethereum. According to the Ethereum whitepaper [15], gas measures the computational resources consumed by transactions and smart contract executions, with each operation requiring a specific amount of gas paid in the network's native cryptocurrency. Every transaction defines a *STARTGAS* value, which limits the maximum number of computational steps allowed, and a *GASPRICE* value, which specifies the fee per step. This mechanism prevents excessive resource consumption and infinite loops, forming a core part of Ethereum's anti-denial-of-service protections. Gas fees significantly influence

the efficiency and usability of decentralized applications, as high costs may discourage user interaction. For this reason, developers apply techniques such as optimized data structures and efficient code design to reduce gas usage. Additionally, Ethereum provides mechanisms like *SELFDESTRUCT*, which removes a contract's code and state from the blockchain to free storage. While useful for outdated contracts, this operation risks permanently deleting state data, potentially affecting users who rely on it. Execution must also remain within gas limits, as exceeding them results in "Out of Gas" errors and failed transactions. [13], [15]

4) *Wallets*: Cryptocurrency wallets are software or hardware tools that allow users to generate, store, and manage the cryptographic key pairs required to interact with a blockchain. Wallets do not hold the cryptocurrency itself but maintain the private and public keys needed to authorize transactions recorded on the distributed ledger. Wallets are categorized into: *Software wallets*, including desktop, mobile, and online solutions; *Hardware wallets*, which store keys offline on dedicated physical devices for increased security; and *Paper wallets*, which represent keys printed or written on physical media, provide strong offline security but have a high vulnerability to physical loss. Each wallet type offers different trade-offs in terms of accessibility, security, anonymity, and key-management responsibility, making the choice dependent on the user's operational needs and risk tolerance. [16]

C. Decentralized Applications and Infrastructure

1) *Decentralized Applications (DApps)*: A blockchain-based decentralized application (DApp) is an application that is executed by multiple users over a decentralized network, such as a public blockchain like Ethereum. The core logic and key data of a DApp are managed via smart contracts deployed on the blockchain, which provides decentralization, persistence, auditability, and immutability of operations and data. While DApps may include client-side components (e.g., user interfaces, web or mobile apps) and off-chain servers, their defining characteristic is that key operations (e.g. state changes or token transfers) are handled transparently by smart contracts. [17]

2) *Decentralized Autonomous Organization (DAO)*: In Decentralized Autonomous Organization (DAO), every single aspect of the organization is managed by the participants themselves, and decisions are made through a consensus reached by voting. In DAOs two types of governance mechanisms can be observed: off-chain and onchain governance. While off-chain governance typically employs social approaches to reach a consensus for governance, on-chain governance utilizes coded mechanisms within the platform to achieve consensus. [18]

To participate in governance operations like publishing proposals, casting votes and enacting actions, members need governance tokens. Every DAOs enforces its own policy when acquiring and using tokens, but usually, they are distributed among the members or can be purchased. One token equals one vote, and the more tokens you use, the greater the member voting weight will be. [19]

DAOs can also have a guardian. Usually, they're controlled by a multisig community composed solely of users who act in the best interest of the DAO. Their purpose is to stop malicious proposals. Guardians and multisig primarily exist for security purposes and serve as safe-rails for proposals. These are not executed immediately, some time passes between the proposing and the implementation. How the multisig should reach a decision and what it can effectively do is defined by the DAO. For example, in Aave, canceling proposals can be done but requires a majority consensus within the multisig. Aave guardians can also pause or unpause a pool or an individual reserve. [20], [21]

D. Decentralized Finance (DeFi)

1) *TradFi vs DeFi*: Traditional Finance (TradFi) refers to the established financial system, including commercial banks, investment banks, and other centralized institutions. These institutions have historically played a central role in the global economy but are often associated with higher fees and barriers that can limit accessibility. Decentralized Finance (DeFi), in contrast, is an emerging ecosystem built on blockchain technology. It enables direct peer to peer financial interactions, such as lending, borrowing, and trading, through algorithms and smart contracts, bypassing the need for a central authority. Assets in DeFi are typically cryptocurrencies and all transactions are recorded on transparent, immutable, and distributed ledgers. [1]

2) *Decentralized Exchanges (DEXs)*: A Decentralized Exchange (DEX) is a platform that allows users to trade cryptocurrencies without relying on a central authority or custodian. Unlike Centralized Exchanges (CEX), where the exchange holds user funds and manages orders, DEXs use smart contracts so trading happens directly between user wallets on the blockchain. A DEX is defined as a blockchain application that allows asset exchange at pricing set by a preset formula, often implemented using an Automated Market Maker (AMM) smart contract. This means that the pricing of trades is decided transparently by an algorithm running on-chain, rather than by an order book managed by a company. Most DEXs work through liquidity pools, where users deposit pairs of tokens that other traders swap between. Another key feature is that execution and settlement happen together on-chain, so ownership of assets transfers instantly because funds never leave the blockchain. DEXs also allow easier and faster token listing, giving new projects immediate access to a market without heavy approval processes, while offering more privacy because they do not enforce strict KYC (Know Your Customer) or AML (Anti-Money Laundering) checks. They also increase transparency, as pricing rules and trade history are public and verifiable by anyone. Overall, DEXs offer a more open and trustless environment, while CEXs are often chosen for higher liquidity and convenience. In this context, Curve represents a specialized DEX built around AMM technology, optimized for stable assets and designed to provide highly efficient swaps with low slippage. [3]

3) *Automated Market Makers (AMMs)*: Automated Market Makers (AMMs) are autonomous software agents that facilitate decentralized token trading by determining prices through transparent mathematical formulas encoded in smart contracts. Unlike traditional market makers, AMMs do not use order books or rely on centralized intermediaries. Instead, they operate via liquidity pools, where the token reserves dictate prices through deterministic algorithms. Effectively, AMMs must strike a balance between sufficient liquidity, accurate pricing, and strong risk management, as low liquidity can result in slippage, high volatility, and price distortions. [2]

4) *Liquidity Pools and Impermanent Loss*: Liquidity pools are decentralized collections of funds that allow token trading on decentralized exchanges (DEXs) without using a central order book. There are two types of users: liquidity providers (LPs) and traders. LPs deposit pairs of tokens into smart contracts that manage the pools, while traders can swap tokens directly through these pools. The prices of tokens are determined by the pool's current reserves. LPs earn a share of the transaction fees, proportional to their contribution to the pool, and they can withdraw their funds and earnings at any time. This system provides continuous liquidity for trading while keeping control of assets decentralized. [22] However, liquidity pools are not without risks. One common issue is impermanent loss, which occurs when the value of the tokens a liquidity provider (LP) has deposited changes compared to simply holding them. As the pool automatically rebalances to maintain its token ratio when prices fluctuate, LPs may end up with a lower value than if they had just held the tokens outside the pool. [23]

E. Stablecoins

Stablecoins are one of the most important types of cryptocurrencies. They are digital assets created to keep a stable value, usually by being tied to another asset like the US dollar. Their goal is to provide a reliable way to store value and make payments in the highly volatile crypto market, without needing to convert crypto back into traditional money. Since stablecoins exist directly on the blockchain, they allow fast and global transactions that cannot be reversed. They also offer a good level of pseudonymity, meaning that users make transactions through wallet addresses instead of their real names. People can see the wallet activity on the blockchain, but they do not know who the owner is. Stablecoins are especially useful in places with currency problems, capital controls, or sanctions. The main stablecoins are Tether (USDT), USD Coin (USDC), DAI, and Binance USD (BUSD), with Tether being the largest. In 2024, stablecoins reached a market value of over \$150 billion, drawing interest from institutions but also raising regulatory concerns because of their connection to traditional finance. [24]

Stablecoins play a central role in Curve Finance. The platform is designed for efficient trading between assets that maintain a similar value, making it especially suitable for stablecoin swaps. Through the StableSwap algorithm, Curve reduces slippage and limits impermanent loss, improving liq-

uidity and trading stability. These aspects will be explored further in section V.

III. GOVERNANCE BACKGROUND OF CURVE FINANCE

A. Curve's Original Vision and Team

Curve is an automated market maker (AMM) created by Michael Egorov in 2020. Egorov wanted to create an AMM with the focus of minimizing slippage when swapping tokens in a pool, particularly for stablecoins.

In order to optimize slippage, Curve uses a combination of both constant product and constant sum functions as its price function. This idea was inspired by an already existing automated market maker called Uniswap, which uses only a constant product formula for the swap of tokens [25].

B. Curve Governance and veCRV Model

This paragraph will focus on the governance model adopted in Curve, its functioning, and the veCRV model, explaining how Curve works at a governance level. Curve operates in the context of decentralized finance (DeFi) and DAOs.

In Curve DAO, smart contracts are connected through Aragon and the veToken model is adopted. Aragon is a DAO-as-a-service platform that offers organizational templates that can be customized and extended, if necessary, by writing applications that can be concatenated with smart contracts [26].

Curve tokens are called CRV and are used for both governance and value accrual. The acronym 've' in veCRV stands for Vote-Escrowed. Vote-Escrowed token models the weight of the vote proportional to the lock time, which replaces the classic one token, one vote rule. In order to gain voting weight, the member must lock their tokens. When locking a CRV token, it becomes a veCRV. The minimum timelock time is 1 week, the maximum timelock is 4 years. The weight of a vote can easily be calculated with this formula:

$$w = a * \frac{t}{t_{max}}$$

a is the number of tokens locked, t is the time locking duration (the time for which they are being locked) and t_{max} is the maximum locking time. Curve DAO votes can be defined as both amount-weighted (the more tokens voted, the greater the weight) and time-weighted (the more the time, the greater the weight). Once CRV is locked, the user can allocate its voting weight between the gauges, increase tokens locked, or withdraw them if their locking time expires. The gauges track the liquidity provided by the users for the liquidity pool and considering the voting weight determine the number of new CRV tokens that will be emitted. Curve gauges are assigned to pools and are the base mechanism for the inflation incentive. Users, in fact, get an inflation reward proportional to his liquidity pool tokens locked. More weight on a certain gauge means more CRV tokens in returns if you invested stablecoins or other in it [7], [8].

C. Emergency Multisig and Guardians

For emergencies, Curve has a multisig called Emergency-DAO composed by 9 members. The EmergencyDAO can intervene if it has a majority of 5 out of 9. Intervenes comprehend: stopping CRV emissions on gauges; recovering ERC20 tokens from the DAO's revenue collection and distribution process; pausing the Peg Stabilization Reserve's associated contracts to stop users from depositing or withdrawing crvUSD. [27]

For security and reliability reasons, audits are essential to identify vulnerabilities in governance processes and smart contracts. Audits can be executed from an internal or external perspective. Curve's team often chooses external auditors for their audits, such as Quantstamp or TrailOfBits. These are performed before and after deployment; results are often made public through their official site. Public audits published by Curve's team show that their main priorities are checking the correctness of the code, the quality of the documentation, and the reliability of the tests. These audits also focus on identifying possible attacks, such as draining contract funds, breaking pool invariants, or blocking liquidity providers from accessing their funds. Governance audits can be useful for identifying bribes and abuses of voting power during proposals [9], [13], [28].

IV. PROBLEM STATEMENT & VALUE PROPOSITION

A. Financial Service Provided

Curve Finance provides a DeFi (Decentralized Finance) protocol designed to minimize slippage when swapping pegged assets, particularly stablecoins. Curve is an automated market maker (AMM) that allows traders to swap tokens along a price curve determined by the relative liquidity of tokens within the liquidity pool. [25]

Users can also lock or escrow tokens for a fixed period in exchange for voting weight. The longer the period, the more voting weight is granted. Curve's token is CRV, while its locked form used for voting is known as veCRV. [7]

Curve has expanded its offerings to include crvUSD, a collateralized debt position (CDP) stablecoin that differs from DAI and other overcollateralized stablecoins by utilizing a novel Lending-Liquidating AMM Algorithm (LLAMMA) to manage collateralization. [6]

Derivatives in the Curve Finance ecosystem can be tradable assets or risk profiles inherent to providing liquidity. The most prominent derivative asset is the Liquid Staking Derivative (LSD), which, unlike stablecoins pegged to fiat currency like the US Dollar, is generally pegged to a non-fiat asset such as ETH or SOL. These LSDs, play a crucial role in StableSwap pools where they are held alongside their base asset, or numeraire, to ensure deep liquidity and enable efficient swapping. The central risk these LSDs introduce is the possibility of a depeg, which exposes Liquidity Providers (LPs) to significant loss. [25]

B. Gaps in Traditional Finance

Traditional decentralized exchange mechanisms, such as those used by Uniswap or Bancor, struggle with inherently

volatile markets and often fail to provide price stability and deep liquidity for stablecoins. Specifically, generic constant-product invariants ($x \cdot y = k$) result in enormous price slippage for stable assets, necessitating massive funds to maintain meaningful liquidity. On the flip side, if one for example loads DAI and USDC into Uniswap's liquidity pool, the returns will be tiny. [4]

In the context of lending and borrowing, traditional DeFi platforms (such as MakerDAO or Aave) utilize discrete liquidation mechanisms where a borrower's position is instantaneously liquidated when collateral value drops, potentially resulting in total loss of the position. [6]

Finally, traditional financial markets rely heavily on middlemen, necessitating exchange owners, orderbooks, or human market makers to facilitate trading activities. [4]

C. Competitive Advantages Over Other Protocols

Curve establishes several distinct advantages over other automated market makers and governance models:

- The StableSwap Invariant: Unlike Uniswap's constant-product invariant, Curve utilizes a combination of constant sum and constant product functions; this mechanism allows for trades with minimal price slippage, typically 100 times smaller than standard AMMs. [4]
- Dynamic Peg for Volatile Assets: For assets not strictly pegged to each other, Curve v2 concentrates liquidity based on an internal oracle. This method creates 5 to 10 times higher liquidity than the Uniswap invariant and generates higher profits for liquidity providers. [5]
- Soft Liquidations (LLAMMA): The crvUSD stablecoin introduces a "soft liquidation" mechanism via LLAMMA. In contrast with discrete liquidation mechanisms cited in the previous section, LLAMMA gradually rebalances collateral into stablecoins as prices drop, reducing the risk of 100% loss for the borrower. [6]
- Governance Alignment (veToken): To address the tension between short-term stakeholder gains and long-term organizational success found in simple token-voting models, Curve implemented the Vote-Escrowed (veToken) model; this requires users to lock tokens for extended periods to gain voting weight, aligning their incentives with the protocol's long-term health. [7]
- Fully Autonomous Market Maker: Curve operates as a fully autonomous market-maker, removing the reliance on intermediaries by functioning without exchange owners, orderbooks, or human market makers responsible for trading. [4]

V. CORE MECHANISM DESIGN OF CURVE FINANCE

A. StableSwap AMM Model

[4] StableSwap, or Curve v1, is an automated liquidity provider specialized in stablecoins. The vision behind StableSwap was to offer a Uniswap-like automated exchange but with very low price slippage in order to increase the APR (Annual Percentage Rate) for stablecoin markets.

1) *Invariant Function*: To minimize slippage, Egorov started from this hybrid invariant:

$$\chi D^{n-1} \sum x_i + \prod x_i = \chi D^n + \left(\frac{D}{n}\right)^n$$

This invariant utilizes both constant sums and constant products. x_i in the summation and product represents the amount of the i-th coin in the pool, generalized for n coins. The sum converges to D , which is the invariant representing total liquidity. The product is approximated to $(D/n)^n$, which is the virtual symmetric approximation of the product of all pool liquidity. Egorov tried to achieve the zero-slippage invariant by introducing leverage in the formula represented with χ . By multiplying χ to the constant-sum invariant and adding it to the constant-product, Egorov constructed an invariant that is the constant-product when the leverage χ is 0 and the constant-sum when the leverage is ∞ . The leverage χ is defined as follows:

$$\chi = \frac{A \prod x_i}{\left(\frac{D}{n}\right)^n}$$

By substituting χ in the hybrid invariant, Egorov defined the final StableSwap invariant:

$$A n^n \sum x_i + D = A D n^n + \frac{D^{n+1}}{n^n \prod x_i}$$

Before using the StableSwap invariant, you need to load a portfolio of coins and then calculate the value of D for which this equation holds true at all times in order to maintain low price slippage. This can be done by finding a converging solution for D or for a coin x_j when all others are known.

2) *Comparison with Constant-Product AMMs*: In comparison to the Uniswap constant-product, StableSwap is better for stablecoins, since it's designed for assets that are meant to be stable. Uniswap adjusts the prices according to the constant-product invariant. This means that although the price is adjusted, the returns could be very tiny because of the price slippage. This is also true for other volatile, price-unstable AMM like Bancor and Kyber.

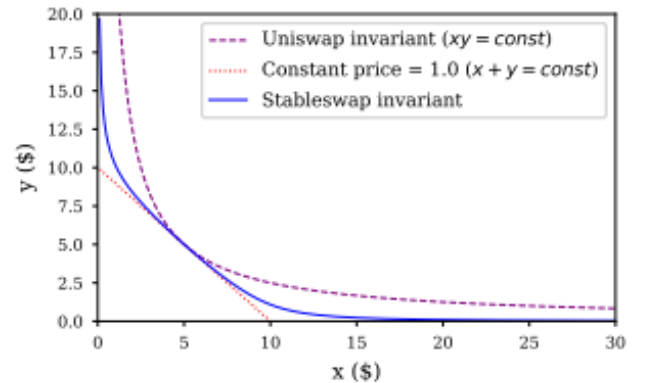


Fig. 1. Price slippage comparison between types of market [4]

B. Cryptoswap AMM Model

[5] After StableSwap, Egorov created CryptoSwap, or Curve v2, with the objective of increasing liquidity for assets and profits compared to Uniswap.

1) *Transformed Pegged Invariants*: In CryptoSwap, all the prices in the price space are transformed by a transformation $T()$. Given $b = (b_0, b_1, \dots)$ the vector of balances in the smart contract, where each b_i represents the balance of the i -th token. Given $p = (p_0, p_1, \dots)$ the set of coefficients for price scales where p_i is the i -th price scale, the real balances b and the transformed balances b' can be converted using these formulas:

$$b = T(b', p) = (b'_0 p_0, b'_1 p_1, \dots)$$

$$b' = T^{-1}(b', p) = \left(\frac{b'_0}{p_0}, \frac{b'_1}{p_1}, \dots\right)$$

The curve of Cryptoswap that Egorov wanted to construct is dependent on deposits in the pools. A constant invariant D for that curve was needed. This is equal to the total deposit in the pool when it is in equilibrium. This invariant is formalized as follows:

$$D = N x_{eq}$$

N is the number of coins and $x_{eq} = (x_{eq}, x_{eq}, \dots)$ the equilibrium point to which the vector p pegs.

Profit and losses can be quantified using a function X_{cp} that quantifies the value of the portfolio without noise and is defined as follows:

$$X_{cp} = \left(\prod \frac{D}{N p_i}\right)^{\frac{1}{N}}$$

X_{cp} is fundamental for repegging prices. CryptoSwap includes a repegging algorithm. All X_{cp} are tracked at every exchange or deposit. Through an internal price oracle, the price vector is updated after each operation. If the adjustment of prices is revealed as inefficient, the last change can be undone since CryptoSwap tracks all losses.

2) *Invariant Function*: A new CryptoSwap invariant was introduced by starting from the StableSwap:

$$K D^{n-1} \sum x_i + \prod x_i = K D^n + \left(\frac{D}{n}\right)^n$$

The leverage K is slightly different and is defined as follow:

$$K = A K_0 \frac{\gamma^2}{(\gamma + 1 - K_0)^2}$$

where K_0 is:

$$K_0 = \frac{\prod x_i N^N}{D^N}$$

A is an amplification coefficient, γ it's a curve shaping parameter that represents the distance between two specific dashed curves that can be seen below in this graphic:

As in StableSwap, this invariant can be solved through Newton's method but needs to be written in the form of $F(x, D) = 0$ and to use specific initial value to have a correct convergence.

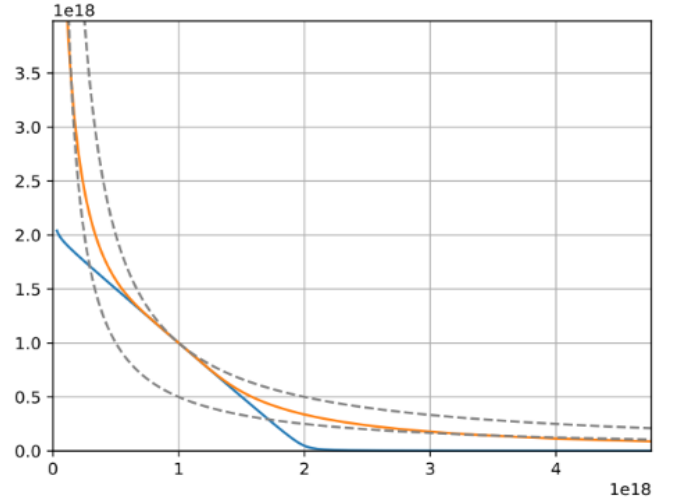


Fig. 2. Comparison of AMM invariants: constant-product (dashed line), stableswap (blue) and from this work (orange) [5]

3) *Dynamic Fees*: In CryptoSwap, the fees are dynamic and are defined within a range set by two variables: f_{mid} , the fees when in equilibrium and f_{out} , the fees in extreme imbalance. Using these, the fee f can be calculated as follows:

$$f = g \cdot f_{mid} + (1 - g) \cdot f_{out}$$

g in the fee's formula is defined as follows:

$$g = \frac{\gamma_{fee}}{\gamma_{fee} + 1 - \left(\frac{\prod x_i}{(\sum x_i / N)^N}\right)}$$

C. Market Structure

Curve's market structure is divided into two main pool types: StableSwap pools and CryptoSwap pools.

1) *Stable Pools*: StableSwap pools use the Curve v1 StableSwap invariant, which enables exchanges between two or more tokens. According to the Curve documentation [29], stable pools fall into three categories:

- **Plain pools** pair two or more stablecoins with each other. One of the main examples is the 3pool, where users can interact with DAI, USDC and USDT.
- **Lending pools** two or more wrapped tokens are paired against each other leaving the backbone operations to other protocol. Tokens in these pools are deposited into external lending protocols, such as Aave or Compound, and therefore exist in their wrapped forms. The Aave pool uses Aave for lending and its wrapped tokens are called aTokens, such as aDAI. The Compound pool works similarly but uses cTokens, such as cDAI. In aTokens, interest accrues as the token balance increases, while in cTokens interest accrues as the exchange rate increases.
- **Meta pools** pair stablecoins with liquidity pool (LP) tokens from other pools. A liquidity provider can deposit DAI, USDC or USDT and receive 3CRV in return. An example is the FRAX meta pool, which allows trading

FRAX for 3CRV, which can then be exchanged for DAI, USDC or USDT in the 3pool [30].

2) *Crypto Pools*: In CryptoSwap pools, the CryptoSwap invariant is used. According to the CryptoSwap documentation [31], these pools focus on swaps between two or more volatile, non-pegged assets such as Bitcoin or Ethereum. They are created by the CryptoSwap Factory and accept two or more assets depending on the type of pool adopted. CryptoSwap pools have two assets and an example is the WETH/WBTC pool.

TriCrypto expands this model by supporting three assets, enabling combinations such as two volatile assets and one stablecoin; for example, WETH, WBTC, and USD. Two further pool types, TwoCrypto-NG and TriCrypto-NG, provide enhanced versions of CryptoSwap and TriCrypto.

3) *Assets and Whitelisting*: Assets include all tokens accepted on Curve and can be grouped into trading tokens, liquidity pool tokens and protocol governance tokens. Trading tokens are supplied by liquidity providers and are exchanged by users. These tokens form the pairs that generate liquidity pool tokens. Some trading tokens also serve as governance tokens, used for decision-making within the protocol [32].

According to the Curve-Assets repository [33], anyone can create pools on Curve. The repository contains the images of all tokens available for use. Curve's native assets are CRV, crvUSD and scrvUSD. As explained in Section III, CRV is the main DAO governance token, crvUSD is Curve's primary stablecoin and scrvUSD is an interest-bearing stablecoin that generates yield simply by holding the token [34].

Curve also includes a system called the MetaRegistry [35], which acts as an aggregator for pool registries. It provides an API that allows users to search for pools containing specific coins and map all assets present in pools.

In earlier versions of Curve, a whitelist controlled which smart contracts could access governance and veCRV. This was enforced by the SmartWalletChecker, which verified whether a contract had been approved by the DAO and blocked non-approved addresses. Following a governance proposal, the majority voted to remove the whitelist, and all smart contracts were allowed thereafter [36].

D. Risk Parameters

1) *LLAMMA*: Lending Liquidating Automated Market Maker Algorithm, or short, LLAMMA is Curve's market-making contract that rebalances the collateral of a loan and is responsible for liquidating and deliquidating collaterals based on market conditions through arbitrage traders. [37]

The core idea of Egorov behind LLAMMA was to implement automatic conversion between collaterals and stablecoins. If the collaterals are high and a user has all in ETH, as the value decreases, they'll be converted to stablecoins. The same happens in the other direction. This is done to reduce loss in liquidation and is only possible with the use of an external price oracle. [38]

In other AMMs the scenario can be described by graphic 3.

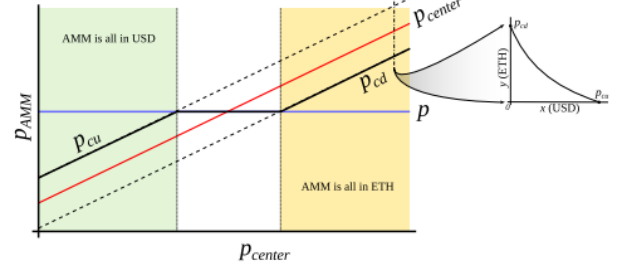


Fig. 3. Usual AMM implemented in markets [38]

As seen in the whitepaper [38] and from graphic 3, from a mathematical point of view: p_{cd} is the current down price, p_{cu} is the current up price, p_o is the oracle price, p_{center} is the price at which liquidity was formed and lastly p is the current price. In this graphic, prices are in the interval $p_{cd} < p_{center} < p_{cu}$, and when p is out of range, the AMM is either all in stablecoin if it goes over p_{cu} , or all in collaterals when over p_{cd} . In other AMMs the scenario can be described by graphic 3.

What LLAMMA wanted to achieve in economic terms can be described in graphic 4. The functions p_{cd} and p_{cu} are

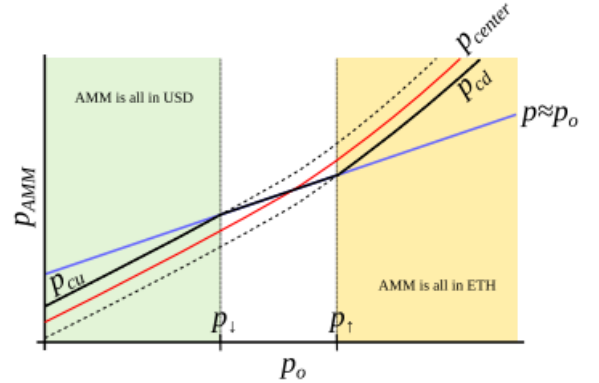


Fig. 4. AMM LLAMMA implemented [38]

functions of p_o such that when p_o grows, they grow even faster. This enables a particular scenario: when ETH is expensive, the AMM will be entirely in ETH, and in USD when ETH is cheap.

E. Liquidation rules

In order to implement this, the price space is split into n bands that touch each other so that if we have a base price p_{base} , we have:

$$p_{\uparrow}(n) = \left(\frac{A-1}{A}\right)^N p_{base} \quad p_{\downarrow}(n) = \left(\frac{A-1}{A}\right)^{N+1} p_{base}$$

The intervals p_{cd} and p_{cu} are defined as follows:

$$p_{cd} = \frac{p_o^3}{p_{\uparrow}^2} \quad p_{cu} = \frac{p_o^3}{p_{\downarrow}^2}$$

It's important to point out that LLAMMA uses p_o defined as ETH/USD price as a price source and that stablecoins could be traded when under the peg $p_s < 1$ (ETH/USD ; 1\$) or when over the peg $p_s > 1$. Know that if $p_s < 1$, the price p in LLAMMA is greater than p_o . If $p = p_o/p_s$ is an approximation of the price, then the oracle price p'_o would be equal to

$$p'_o = p_o \sqrt{p_s}$$

The stablecoins obtained would be higher by $1/p_s$ if $p_s < 1$, that's why it's not very good to be over the peg ($p_s > 1$) for long periods of time.

F. Caps and Stabilizer

For this matter, an automatic stabilizer has been implemented. Given p_s the peg price, and d_{st} the debt, which are the stablecoins minted to the stabilizer, and a function f_{dx} which does exactly as `get_dx` in the stableswap AMM, then we can keep the reserves stabilized using a slow stabilizing mechanism which uses the peg price and the borrow r :

$$p_s = \frac{f_{dx}(d_{st})}{d_{st}} \quad r = r_0 * 2^{-\frac{p-1}{h}}$$

In the borrow r , h is the change in p_s at which the rate r changes by factor of 2. d_{st} will be equilibrated depending on the rate $p_s = 1r_0$. If r_0 is reduced, the prices decreases and the the system is forced to burn the debt d_{st} inducing borrowers to borrow and dump stablecoins. If it is increased, the system is forced to increase the debt stabilizing deposits and inducing borrowers to return loans in order to push p_s up.

G. Multi-chain Deployment

Curve products and assets are deployed across multiple blockchains. Each network offers unique advantages in terms of transaction speed and costs, providing various choices to users. [39]

1) *Ethereum*: Curve was first deployed on Ethereum, and as of today, it continues to be Curve's primary network. Some components, such as `crvUSD` minting and DAO governance, are only available on the Ethereum network. Currently, there is no plan to deploy them to other networks. To use Curve on alternate networks, you'll typically need to bridge your funds from Ethereum using the target network's bridge. Curve's DEX and assets are available on many chains, thanks to the bridge system. The same goes for Curve Lending, which is deployed on Ethereum and selected L2s (Arbitrum, Frax, Optimism); it is scheduled to be expanded to new networks soon. Network availability can be checked on Curve Swap product page. CRV rewards are available on all chains that have active gauges. New markets must be added to the GaugeController to receive rewards. [39]

2) *L2s (Arbitrum, Base, Fractal, Optimism)*: Arbitrum, Base, Frax and Optimism are L2s or level-two chains. On L2 chains, you can: swap tokens, provide or remove liquidity from pools and stake on L2 gauges. Users can earn CRV rewards through Llamalend with a delay of 1 week. Also, while voting is exclusive to the Ethereum network, after a successful vote,

actions executed are broadcasted to L2 networks thanks to Curve's cross chain governance system `x-gov`. [40]

3) *Sidechains (Polygon)*: Lastly, chains such as Polygon, are called Sidechains. These are separated and independent blockchains which are connected to a main chain with the purpose of improving scaling and interoperability. On sidechains, you can only swap tokens, stake on gauges and provide liquidity. [41]

VI. USER EXPERIENCE AND TUTORIAL

A. Wallet Connection Steps

1) *MetaMask*: To interact with Curve Finance, the user must first connect a cryptocurrency wallet. One of the most popular choices is *MetaMask*, a crypto wallet available as a browser extension and mobile app. MetaMask allows users to store tokens, approve transactions, and connect to decentralized applications (dApps).

After installing the extension and creating or importing an account, the user can open Curve Finance and select *Connect Wallet* → *MetaMask*. MetaMask will request approval to link the wallet to Curve.

Curve operates on various blockchains, such as Ethereum, Arbitrum, Polygon, Optimism, and Avalanche. For this reason, the user must also select the correct *network* inside MetaMask. A blockchain network determines where the transaction is executed, which tokens are available, and the gas fees required. Selecting the correct network is therefore essential for successful swaps or liquidity operations.

B. Example User Flow

1) *Executing a Swap*: Curve Finance allows users to swap tokens directly through its interface. Before swapping, the user must already hold cryptocurrency in their MetaMask wallet. If not, tokens can be purchased on centralized exchanges such as Binance or Coinbase and then transferred to the wallet.

Once the user has funds available, the swap can be executed as follows:

- 1) Select the token to exchange (*Token A*).
- 2) Select the token to receive (*Token B*).
- 3) Choose the amount to swap.

During this process, Curve displays important information to help the user evaluate the trade:

- **Exchange Rate**: shows how much of Token B is received for Token A, with fees included.
- **Price Impact**: indicates how much the trade influences the pool price (large swaps have higher impact).
- **Trade routed through**: in some cases, swaps may pass through multiple pools to ensure the best price.
- **Estimated TX Cost**: represents the network transaction fee, which varies based on blockchain congestion.
- **Slippage Tolerance**: the maximum acceptable price variation during the swap. This value can be adjusted by the user.

If it is the user's first time interacting with a specific token or smart contract, an *approval transaction* is required. After

approving, the user can confirm the swap and execute it on-chain.

2) *Providing Liquidity*: Instead of swapping, users can also provide liquidity to Curve pools to earn passive yield from trading fees. To become a liquidity provider (LP), the user must follow these steps:

- 1) Navigate to the *Pools* section in Curve Finance.
- 2) Select the desired pool.
- 3) Deposit one or more tokens that the pool supports.

Most Curve pools contain stablecoins (such as DAI, USDC, USDT), but some include assets with similar price behavior, such as ETH–stETH. These pools help reduce volatility and minimize risks like impermanent loss.

Each pool displays its details:

- **Base vAPY**: variable APY generated from trading fees only.
- **tAPR**: total APR including Base vAPY + CRV rewards + external incentives.
- **CRV Rewards**: Curve’s native token distributed to liquidity providers as an incentive. It can be claimed or staked to increase returns.

When adding liquidity, the user interface also shows:

- **Minimum LP Tokens Received**: tokens representing the user’s share of the pool. They are required later to withdraw assets.
- **Slippage Estimate Bonus**: indicates if the deposit improves or worsens the pool balance (affecting the amount of LP tokens received).
- **Transaction TX Cost**: gas fees needed to execute the deposit on the blockchain.
- **Adjustable Slippage Tolerance**: maximum deviation in pool price accepted during the deposit. A lower value protects the user from bad pricing but may cause the transaction to fail; a higher value increases execution success but with more risk.

After depositing, the liquidity provider receives LP tokens in their wallet. These tokens can also be *staked* to earn additional CRV or other incentive rewards.

Users can later:

- **Withdraw liquidity** by returning LP tokens to redeem their assets.
- **Unstake rewards** to collect earned CRV or incentives.

A user can also exchange assets within the same pool, benefiting from lower slippage compared to standard swaps.

C. Screenshots

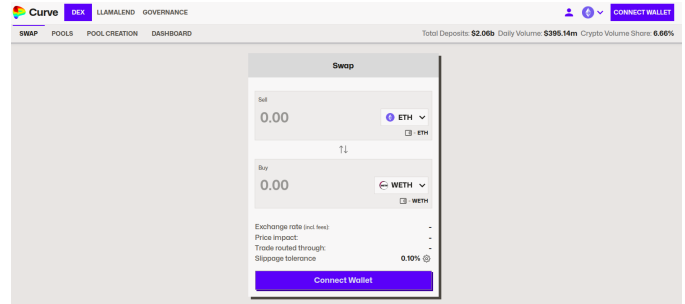


Fig. 5. Curve Finance homepage showing the main interface and navigation.

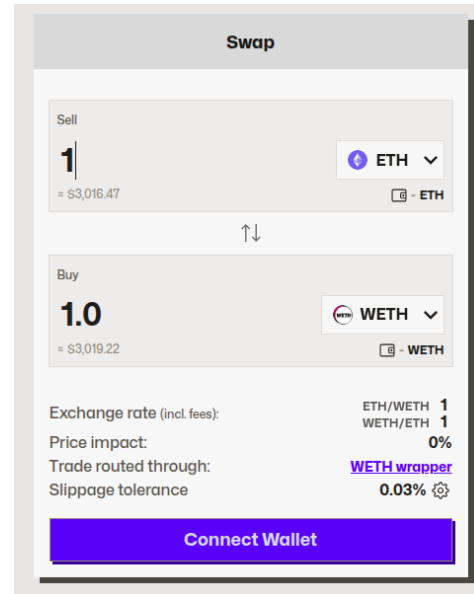


Fig. 6. Example of a token swap on Curve Finance, including input/output selection and estimated transaction details.

Curve

SWAP

LLMALEND

GOVERNANCE

CONNECT

WALLET

SWAP

POOLS

POOL CREATION

DASHBOARD

Total Deposits: \$2,085

Daily Volume: \$395.14m

Crypto Volume Share: 6.66%

Q

Search by tokens or address

ALL CRY

USD

BTC

ETH

CRVUSD

TRICRYPTO

CRYPTO

STABLE NO

Sort by Volume (desc)

Hide very small pools

Pool	Base vAPY (%)	Rewards tAPR (%) CRV + Incentives	Volume	TVL
Spark.fi USDT Reserve eUSDT USDT 2.91% \$104.82m \$50.02m				
Spark.fi PYUSD Reserve PYUSD USDC 0.96% \$78.50m \$102m				
Curve.fi Strategic USD Re.. USDT USDT 3.18% 3.26% + 8.16% CRV \$54.84m \$3.55m				
3pool DAI USDC USDT 0% 0.0000981587% + 0.00046397% CRV \$28.44m \$168.02m				
crvUSD/USDT USDT ecrvUSD 0.93% 5.90% + 14.75% CRV \$16.86m \$69.95m				
crvUSD/USDC USDC ecrvUSD 1.01% 5.54% + 13.85% CRV \$13.07m \$38.04m				
YB WBTC ecvUSD WBTC 16.02% \$7.02m \$98.25m				
YB WBTC ecvUSD WBTC 13.87% \$6.14m \$97.87m				

Fig. 7. List of available liquidity pools, showing the assets contained in each pool and their respective rewards.

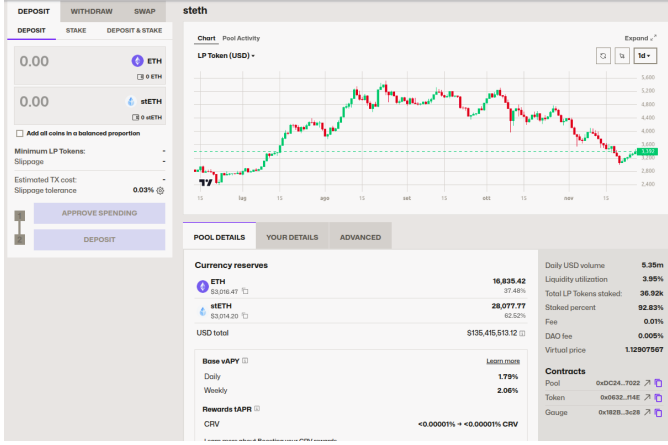


Fig. 8. Dashboard of a specific liquidity pool, where users can deposit assets, view rewards, stake LP tokens, withdraw liquidity, and perform swaps within the pool.

VII. TOKENOMICS AND INCENTIVES

A. CRV Token Roles

1) *Governance*: As noted in Section III-B, CRV is the governance token of the Curve DAO. Users can lock CRV to get veCRV (vote-escrowed CRV), which gives them voting power in the DAO. With veCRV, users can vote on protocol settings, new pool approvals, and how token emissions are distributed.

2) *Utility*: Curve’s liquidity gauges send CRV inflation to liquidity providers based on the votes of veCRV holders through the GaugeController and Minter contracts. This system motivates users to participate in governance, as veCRV holders can choose how emissions are assigned to approved gauges. Gauges track liquidity over time using LP token deposits and make sure CRV is distributed fairly based on the liquidity provided, with the possibility of extra rewards. [42]

3) *Fee Distribution*: Curve’s fee distribution system collects admin fees from liquidity pools and interest from crvUSD loans in the FeeCollector. These funds are swapped into a single token like crvUSD using the CoWSwapBurner, then sent to the FeeDistributor. The FeeDistributor sends these rewards to veCRV holders based on their share, and users can claim them through the claim function. The FeeSplitter divides crvUSD fees among different controllers in one transaction. This system works on Ethereum mainnet (with two FeeDistributors for both 3CRV and crvUSD) and also on sidechains through bridging, using Hooker for transfers. [43]

B. Incentives

1) *Liquidity Mining*: Liquidity mining on Curve distributes CRV emissions to users who stake LP tokens in pool gauges. Rewards are proportional to the user’s share of the total liquidity. The system uses integrals to track balances over time, taking into account deposits, withdrawals, and changes in inflation rates, gauge weights, or epochs. Users claim rewards based on `integrate_fraction` values, which lets the protocol

allocate emissions accurately without constant manual updates. [42]

2) *Gauge Weights*: Gauge weights decide how much CRV each gauge receives and are set by veCRV holder votes. The GaugeController applies these weights weekly. Votes create scheduled changes with bias and slope values, and each user’s slope is tracked to reflect the linear decrease of voting power over time. Because changes only take effect at the start of each epoch, gauges can only be updated once every 10 days to keep the system stable. [42]

3) *veCRV Boosting Mechanism*: The veCRV boosting system increases a user’s effective liquidity balance b_u^* for rewards by up to 2.5x. It combines 40% of the user’s actual balance with a veCRV-based share of the pool’s total working supply S . The formula is:

$$b_u^* = \min \left(0.4b_u + 0.6S \frac{w_i}{W}, b_u \right),$$

where w_i is the user’s veCRV voting weight for that gauge and W is the total veCRV weight in the gauge. The boost cannot exceed the user’s real balance. Boosts are activated when users deposit, withdraw, or manually update, and they stay active until the next action. If a user’s veCRV lock expires, other users can “kick” them to reset the boost, encouraging users to keep their locks active. [42]

C. Emission Schedule & Sustainability

CRV uses a piecewise linear inflation schedule starting at 22% per year (around 2 million CRV per day at the beginning), with a yearly decrease of about 15.9% as new epochs start. The first 1.273 billion tokens (42% of the total ≈ 3.03 billion supply) vest over time, and all inflation goes to gauges based on their weights and liquidity levels. Pools that become inactive receive fewer emissions, promoting ongoing activity and long-term sustainability. [42]

D. Treasury and Revenue Model

The Curve DAO earns revenue mainly from trading fees paid by users swapping tokens in liquidity pools and from interest produced by crvUSD lending markets. These fees accumulate in the FeeCollector as different tokens. They are then swapped into crvUSD through CoWSwap on Ethereum and Gnosis Chain, allowing flexible and efficient conversions without fixed token paths. The swapped crvUSD is sent to veCRV holders as rewards and also helps cover protocol costs such as development. Sidechains process fees using local burners or bridge them back to Ethereum. Planned upgrades on Arbitrum aim to improve cross-chain efficiency. This model keeps revenue sustainable by connecting it directly to platform usage. [43]

VIII. RISK ANALYSIS

A. Technical Risks

One of the main technical risks involves the fact that Curve cannot easily patch newly uncovered bugs without having to fully redeploy the smart contracts. [6] This architectural rigidity significantly amplifies the impact of the Stableswap

algorithm's known computational inefficiencies. Specifically, the algorithm suffers from high gas utilization in multi-asset pools compared to newer competitors like Wombat. Because the contracts cannot be easily upgraded, these performance bottlenecks are effectively entrenched, meaning they cannot be resolved without a complex and risky migration of liquidity to entirely new contracts. [44]

A further technical concern involves the integrity of the governance contracts. An audit revealed that Curve's DAO utilizes an Aragon voting framework containing known vulnerabilities, which exposes the protocol to risks such as the manipulation of voting power and susceptibility to spam attacks. [45]

One last technical issue involves the fact that crvUSD's LLAMMA mechanism relies heavily on external price oracles. To work, it must pull data from a combination of oracles (including Uniswap TWAP, Chainlink, and Tricrypto) and apply an exponential moving average (EMA) to reduce volatility and limit exposure to manipulation risk. However, the LLAMMA system still relies on external price oracles that could be subject to manipulation. [6]

B. Economic risks

Liquidity Providers (LPs) on Curve are exposed to two main economic risks: the market risk of a token depegging, and the impermanent loss incurred by the pool if a token loses its peg. In the event of a depeg, LPs are highly exposed to these losses and exhibit strong negative convexity, acting effectively as option sellers betting that prices will not move persistently in one direction. During depegging events (such as the UST crash), Curve's constant product mechanism causes the LP to accumulate large quantities of the depegged token while selling off the remaining tokens. Importantly, impermanent loss is mostly related to pools containing volatile, non-stablecoin assets, because the relative price movements between these tokens are larger and more frequent. In contrast, pools of stablecoins or other assets with tightly correlated values experience minimal impermanent loss, since their relative prices remain close to 1:1, limiting the rebalancing losses incurred by LPs. [25]

One further concern may arise from the fact that, as collateral positions are converted into LP positions under LLAMMA, depositors are exposed to market-making losses due to constant rebalancing. This implies the potential for increased collateral value loss during periods of extreme price volatility and high gas fees, which can impede the active rebalancing process. [6] Nevertheless, the Curve Stablecoin design aims to mitigate these losses: simulations indicate that if the collateral price falls 10% below the liquidation threshold and recovers within 3 days, only 1% of the collateral is lost, compared to a total liquidation. [24]

One last economic risk involves the fact that the LLAMMA system relies heavily on arbitrageurs to rebalance pools and maintain the crvUSD peg. This means that if their intervention is delayed, for instance, due to high gas fees, it can lead to amplified value losses. [6]

C. Governance Risks

A primary governance concern arises from the fact that Convex Finance controls approximately 45% of all locked CRV (veCRV), effectively establishing it as the dominant voter within the Curve ecosystem. This concentration of influence is further evidenced by analysis showing that a select few delegates often control vast amounts of voting rights, thereby amplifying the risks associated with centralization. [7], [45]

Analysis of the veToken model further underscores these centralization concerns by demonstrating that votes follow bribes, with a 0.99 correlation observed between the percentage of bribes distributed and the share of votes received for a gauge. Higher-layer protocols like Votium exert immense influence within this dynamic; for instance, Frax Finance, despite directly holding less than 0.1% of veCRV, commands significant sway by leveraging Votium to pay over one hundred million dollars in bribes. [7]

These systemic vulnerabilities were starkly illustrated by the governance attack orchestrated by Mochi Inu, where the project created a pool and utilized Mochi tokens to artificially inflate rewards, thereby seizing disproportionate influence over governance. This incident highlights the persistent risk that malicious proposals could be successfully passed through vote buying or the sudden activation of dormant whales, posing a continued threat to the protocol's integrity. [45]

Finally, while gauge proposals (which determine the allocation of CRV emissions and financial rewards) attract significant participation due to economic incentives like bribes, non-gauge proposals (such as administrative decisions) lack these direct financial rewards. This lack of incentive leads to drastically lower voter turnout, a phenomenon described as Voter Apathy. [7]

D. Safety Mechanisms

To mitigate these vulnerabilities, Curve employs specific safeguards, most notably an Emergency DAO empowered to intervene during critical events. This mechanism proved vital during the Mochi Inu attack, where the Emergency DAO effectively neutralized the attack by stripping the malicious pool of its rewards. Furthermore, robust protocols often include an Emergency Shutdown feature as a measure of last resort, granting token holders the ability to reclaim a proportional share of the treasury to prevent total loss in catastrophic scenarios. [45]

To maintain the peg of crvUSD, Curve uses "PegKeepers" who can mint/burn uncollateralized crvUSD and deposit or withdraw it from the stableswap pool when the price deviates from one dollar, stabilizing the price without huge capital reserves. [6]

In addition to these active intervention mechanisms, the protocol utilizes automated economic levers to enforce stability and protect participants. The system implements a dynamic monetary policy wherein borrowing rates for crvUSD automatically adapt to the variance of the token's price relative to its peg; specifically, interest rates increase if the price falls below one dollar to incentivize loan repayment. [6]

IX. USAGE & PERFORMANCE METRICS

A. Scale & Activity

To evaluate the operational scale of Curve Finance, we analyze three core dimensions of on-chain activity. These indicators collectively describe the protocol’s capacity to attract liquidity, facilitate economic activity, and maintain a consistent user base.

Total Value Locked (TVL) represents the aggregate value of assets deposited in Curve’s liquidity pools across supported chains. It is commonly used as a proxy for a protocol’s scale, capital efficiency, and perceived safety. A sustained or expanding TVL indicates strong liquidity provider (LP), confidence and competitive yield conditions, whereas contractions may reflect shifts in market sentiment, alternative yield opportunities, or security-related events. [46]

Daily trading volume quantifies the economic activity occurring within Curve pools and is directly linked to fee generation and liquidity provider returns. High volume relative to TVL is often interpreted as a positive indicator of capital efficiency. [47]

Finally, user engagement is assessed through the number of unique active addresses interacting with Curve contracts, as well as the frequency of their transactional behavior. A stable base of daily or monthly active addresses suggests sustained organic demand, while significant fluctuations may indicate periods of speculative activity, migration to alternative platforms, or changes in incentive programs. [48]

TABLE I
COMPARISON OF USAGE AND ACTIVITY METRICS AMONG CURVE
FINANCE AND SELECTED DEX COMPETITORS

Protocol	TVL	D. Trading Volume	D. Active Users
Curve Finance	\$2,133b	\$97.17m	2.5k
Uniswap	\$4.16b	\$1,433b	6.1m
SushiSwap	\$107.58m	\$797.5k	11.9k
Balancer	\$256.91m	\$12.46m	7.1k

B. Economics

The economic sustainability of Curve Finance can be evaluated through three interrelated dimensions. Together, the following indicators provide insight into the efficiency of Curve’s incentive structure, the resilience of its business model, and the long-term viability of its liquidity provisioning framework.

Curve Finance generates revenue primarily from trading fees collected across its AMM pools. These fees accrue either to LPs or, depending on pool configurations, partially to the protocol treasury. Revenue from fees reflects the underlying utility of Curve’s specialized stable-swap architecture and serves as a proxy for the protocol’s ability to capture value from on-chain trading activity. [49], [50]

Another central element of Curve’s economic design is its token-based incentive system, which distributes CRV emissions to liquidity providers. These emissions are intended to attract and retain deep liquidity, especially in strategic pools, but they also represent a cost borne by the protocol. The magnitude of this cost depends on the issuance rate of CRV,

the market price of the token, and the distribution of boosted rewards governed by the veCRV mechanism. Excessive reliance on emissions can depress token value and introduce long-term inflationary pressure, whereas calibrated emissions may effectively bootstrap liquidity and stabilize pool depth. [51]

Finally, profitability can be approximated by comparing the protocol’s fee revenue with its emissions expenditures. While most of Curve’s fee revenue is distributed directly to LPs rather than accruing to the protocol treasury, profitability at the system level can still be analyzed through a net value creation lens:

$$\text{Net Economic Output} = \text{Fee Revenue} - \text{Emissions Cost.}$$

If fee generation sustainably offsets or exceeds the opportunity cost of emissions, the protocol can be considered economically efficient. [52]

TABLE II
COMPARISON OF ANNUALIZED ECONOMIC METRICS ACROSS SELECTED
DECENTRALIZED EXCHANGES

Protocol	Revenue p.a.	Incentives p.a.	Earnings p.a.
Curve Finance	\$12.5m	\$49.36m	-36.86m
Uniswap	—	—	—
SushiSwap	\$10.45m	—	—
Balancer	\$3.16m	—	3.16m

C. Reliability

In this last section we assess the reliability of the protocol by considering security audits, historical security incidents, and operational downtime.

Curve Finance has undergone multiple audits conducted by reputable firms such as Trail of Bits, Quantstamp, MixBytes, and ChainSecurity. These assessments primarily focus on the AMM math, pool invariants, governance contracts, and veCRV token mechanics. While comprehensive, these audits emphasize that Curve’s architecture is highly specialized; thus, audit coverage is extensive but narrowly focused on Curve’s tailored bonding-curve implementations. Continued audits and formal verification efforts remain essential given the constant deployment of new pool types and extensions. [28]

Curve Finance has experienced relatively few critical security incidents throughout its operational history, with the most significant occurring on 30 July 2023, when several of its liquidity pools were exploited due to a vulnerability in specific versions of the Vyper compiler rather than in Curve’s own smart-contract logic. The incident affected multiple pools, including aETH/ETH, pETH/ETH, msETH/ETH, and CRV/ETH, and resulted in multi-million-dollar losses, initially estimated at approximately \$70 million, before partial fund recovery by white-hat actors and MEV bots. The exploit stemmed from a failure in the compiler’s reentrancy-lock enforcement, enabling unauthorized withdrawals from vulnerable pools. In addition, in the immediate aftermath (early August 2023), Curve coordinated with auditors and

ecosystem partners to freeze affected pools, patch compiler-related issues, and implement enhanced validation of external dependencies. Beyond this event, Curve has not recorded additional major protocol-level security breaches, underscoring a generally strong security posture. [53]

Finally, since the core smart contracts operate without centralized servers, they do not experience conventional downtime. However, auxiliary components such as Curve’s user interface, APIs, and subgraph indexing services may experience intermittent availability issues. These incidents do not affect vault solvency or on-chain AMM operations but can temporarily hinder user interaction. Overall, Curve demonstrates high operational reliability consistent with on-chain automated market-making infrastructure.

X. NOTABLE INCIDENTS & POST-MORTEMES

A. 2023 Reentrancy Exploit

On July 30th 2023, Curve Finance experienced losses due to one of the biggest attacks on the Curve blockchain. This reentrancy attack caused Curve to lose 68 million dollars. This genre of attack is a vulnerability exploit where unsynchronized states are leveraged during external contract calls, leading to repeated executions of actions that would normally have been performed only once. The main focus of this attack is the withdrawal of funds. [54] In order to succeed, hackers carefully studied the DeFi blockchain structure and proceeded to use a combination of tactics based on price manipulation and flash loans. By exploiting vulnerabilities found in the Vyper compiler, the prices of the stablecoins were altered on Curve’s market and later traded for other tokens at inflated rates. By utilizing flash loans, they borrowed cryptocurrency without providing collateral, as it would be repaid in the same transaction. As in every attack, measures for concealment of identity and anonymity were adopted by the attackers. When the vulnerability was exposed, Curve also experienced a drop in popularity and CRV price. After one month and an offer not to take legal action, most of the coins were returned, but only 70% of them. In the end, the loss settled at 20\$ million. [55]

B. Governance and Incentive attacks

According to [9], other incidents occurred on July 20 and in November 2021. In July 2020, a security audit found a voting contract with code vulnerabilities that allowed users to abuse voting power and enabled potential spam attacks. In November 2021, Mochi Inu set up a Curve pool with the intention of attracting funds that could be used to increase rewards for the Mochi pool, all to gain influence over Curve’s governance. This incident was stopped thanks to EmergencyDAO, which cut off the pool’s rewards. Actually, Curve isn’t out of danger because of the Curve Wars. There is strong competition among the DeFi projects to attract rewards to their pools. To do this, protocol bribery is often used to gain more veCRV tokens to redistribute rewards in their pools.

C. Lessons Learned

The majority of Ethereum smart contracts are written in Solidity. This is not the case since Curve’s smart contracts are written and compiled in Vyper. After the 2023 reentrancy attack success, the fault was partially attributed to vulnerabilities in Vyper code. Since then, Vyper developers have worked to develop a patch. In patch 0.3.1, the allocation of unused storage slots was fixed. It was, in fact, this bug that allowed the reentrancy attack; solving this allocation bug meant not assigning new storage slots to reentrancy keys if one was already assigned. [56]

Audits are conducted to discover bugs and security issues; it’s highly recommended to write robust code and to follow best practices in terms of security. For reentrancy attacks, two techniques help with the prevention and mitigation of this genre of attack: Check-Effects-Interactions (CEI) and reentrancy guard locks. CEI is a methodology for developing code that follows three stages:

- Check: before the transaction, balance checks are conducted to prevent an attacker from making repeated withdrawals.
- Interactions: after the checks, the transaction is executed.
- Effects: after the transaction, the internal state is updated.

Reentrancy guard locks are used to ensure that multiple executions of a function at the same time are not allowed. A variable is locked at the start of the function and released at the end of it, neutralizing reentrancy attacks. [57]

XI. CONCLUSION

Curve Finance has established itself as one of the most important protocols in the DeFi ecosystem thanks to its specialized AMM models, strong liquidity, and innovative governance structure. Its StableSwap and CryptoSwap algorithms offer very efficient trading for stablecoins and pegged assets, reducing slippage and improving the experience for both traders and liquidity providers. The introduction of crvUSD and the LLAMMA mechanism also shows Curve’s ability to innovate, offering a new way to manage collateral and reduce liquidation risk through smoother, “soft” liquidations. However, Curve also has several limitations. Its smart-contract structure is difficult to upgrade, which makes bug fixes slow and sometimes risky. Economic risks remain significant, especially when stablecoins lose their peg or when liquidity becomes unbalanced. Governance risks are also notable, since the veCRV model encourages power concentration through token locking, and large external actors like Convex Finance can strongly influence decisions. Past incidents such as the Vyper compiler exploit and governance manipulation attempts, show that Curve is still vulnerable to technical and organizational attacks, even if safety mechanisms like the Emergency DAO help reduce damage. Despite these challenges, Curve continues to be a key piece of DeFi infrastructure. Many other protocols depend on its liquidity, and its pools are often used as reference points for stablecoin pricing. Curve’s multichain expansion and cross-chain governance system also show its intention to

remain relevant in a rapidly evolving industry.

In summary, Curve Finance combines advanced mathematics, innovative tokenomics, and a strong role within the broader DeFi ecosystem. Its strengths make it a leading platform for stable asset trading, while its limitations highlight the need for continuous improvement in security, governance, and risk management. Overall, Curve remains a foundational protocol whose evolution will continue to influence the direction of decentralized finance.

CONTRIBUTION STATEMENT

Each member reviewed and refined all sections of this document written by the other members to ensure accuracy and reduce redundancy. Specifically, each member was responsible for the following parts:

- Simone Roma: Governance Background of Curve Finance, Core Mechanism Design of Curve Finance, Notable Incidents & Post-Mortems.
- Cristian Sirghie: Introduction, Background, User Experience and Tutorial, Tokenomics and Incentives, and Conclusion.
- Davide Pietragalla: Problem Statement & Value Proposition, Risk Analysis, Usage & Performance Metrics.

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